

MTH203: Sequences and series

Lecture 13

1. CAUCHY SEQUENCES

Let X be a metric space with distance function d . Let $(x_n)_{n \in \mathbb{N}}$ be a convergent sequence in X and let $x = \lim_{n \rightarrow \infty} x_n$. Then, for any “sufficiently large” integer n , x_n is close to x . Thus, if m and n are both “sufficiently large” integers, both x_m and x_n are close to x . Thus, they must be close to each other as well.

Now, let us state this rigorously. Let $\epsilon > 0$ be given. Then, there exists N such that for all $n > N$, we have $d(x_n, x) < \epsilon/2$. Thus, if m and n are integers such that $m > N$ and $n > N$, we have $d(x_m, x) < \epsilon/2$ and $d(x_n, x) < \epsilon/2$. Thus, by the triangle inequality, we have

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \epsilon/2 + \epsilon/2 = \epsilon.$$

Thus, we see that there exists an integer N such that if m and n are both greater than N , we have $d(x_m, x_n) < \epsilon$. This leads us to the following definition.

Definition 1 (Cauchy sequence). Let X be a metric space with distance function d . A sequence $(x_n)_{n \in \mathbb{N}}$ is said to be a *Cauchy sequence* if for every $\epsilon > 0$, there exists an integer N such that for any integers n and m satisfying $n \geq N$ and $m \geq N$, we have $d(x_m, x_n) < \epsilon$.

We have proved the following result in the above discussion:

Lemma 2. *Any convergent sequence in a metric space is a Cauchy sequence.*

The converse of this statement is not true in general. It is possible for a Cauchy sequence in a general metric space to not be convergent.

Example 3. Let X be a metric space and let $(x_n)_{n \in \mathbb{N}}$ be a convergent sequence in X . Let $x = \lim_{n \rightarrow \infty} x_n$ and suppose that $x \neq x_n$ for any n . (It is very easy to construct such examples. For instance, take X to be $\mathbb{R}$ and take $(x_n)_{n \in \mathbb{N}}$ to be a sequence of rational numbers converging to some irrational number, which we denote by x .)

Then, let Y be the metric space obtained by removing x from X . In other words, the underlying set of Y is just $X \setminus \{x\}$. The distance function on Y is just the restriction of the distance function on X . The sequence $(x_n)_{n \in \mathbb{N}}$ can be viewed as a sequence in Y since $x_n \neq x$ for any n . Thus, it is a Cauchy sequence in X . Since the distance between x_m and x_n is the same whether we view these points as being in X or Y , it follows that this is also a Cauchy sequence in Y . However, it clearly cannot converge to any point in Y . Indeed, if it converges to some point y of Y , then it must converge to y in X as well. But the limit of the sequence in X is unique and so we must have $x = y$. This is a contradiction.

Definition 4. A metric space X is said to be *complete* if every Cauchy sequence in X converges.

Remark 5. (This remark is not part of the syllabus for this course.) We saw in the example above that the space Y fails to be complete because the limit of a Cauchy sequence was “removed”. This is a fairly typical situation. For any space Y , there exists a complete space $\tilde{Y}$ such that Y is a subset of $\tilde{Y}$ and the distance function on Y is the restriction of the distance function on $\tilde{Y}$. In other words, Y has been created by removing some points from the complete metric space $\tilde{Y}$.

Let us try to construct the complete metric space $\tilde{Y}$ from Y . Suppose Y is not complete. Then, for any Cauchy sequence in Y that does not converge, let us add a new point to Y to serve as the limit of such a sequence. In other words, we forcibly

add the limits that are missing. This can create a bit of a problem because it might be that two Cauchy sequences $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ are “supposed to have the same limit”. Thus, this idea does not really work.

We say that two Cauchy sequences $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ are *equivalent* if for every $\epsilon > 0$, there exists an integer N such that for any $n > N$, we have $d(x_n, y_n) < \epsilon$. One can easily verify that this is an equivalence relation on the set of Cauchy sequences in Y . We can now amend our above construction. For every equivalence class of Cauchy sequences, we add a point which is supposed to serve as the limit of all these sequences. Suppose that the new set (consisting of all the points of Y and the new points that we have added) be denoted by $\tilde{Y}$.

We now define a distance function on $\tilde{Y}$. Note that every point y of $\tilde{Y}$ is the limit of some Cauchy sequence in Y . Indeed, this is obvious if y is one of the new points that have been added. But if $y \in Y$, then it is just the limit of the constant sequence $(y_n)_{n \in \mathbb{N}}$ where $y_n = y$ for all n .

Suppose x and y are two points of $\tilde{Y}$. We need to decide what the distance between these two points should be. Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be sequences in Y which converge to x and y respectively. For every $n \in \mathbb{N}$ let $\alpha_n = d(x_n, y_n)$. Then, for any integers m and n , we have (due to the triangle inequality)

$$d(x_n, y_n) \leq d(x_n, x_m) + d(x_m, y_m) + d(y_m, y_n).$$

Thus,

$$d(x_n, y_n) - d(x_m, y_m) \leq d(x_n, x_m) + d(y_m, y_n).$$

Interchanging the roles of m and n in the above computation, we get the inequality

$$d(x_m, y_m) - d(x_n, y_n) \leq d(x_n, x_m) + d(y_m, y_n).$$

These two inequalities imply that

$$|d(x_n, y_n) - d(x_m, y_m)| \leq d(x_n, x_m) + d(y_m, y_n).$$

This shows that the sequence $(\alpha_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$. We will prove below that $\mathbb{R}$ is a *complete metric space*. Thus, the Cauchy sequence $(\alpha_n)_{n \in \mathbb{N}}$ converges to some real number α . Then, we set the distance between x and y to be α .

It is easy to verify that this defines a distance function on $\tilde{Y}$ and that the restriction of this distance function to the points of Y is the distance function that we started with. One can also easily verify that, with this distance function, $\tilde{Y}$ is a complete metric space. (This has just been a sketch of a proof. For the details, see the reading suggestions below.)

We will make use of the following obvious property of Cauchy sequences:

Lemma 6. *Every Cauchy sequence is bounded.*

Proof. Let X be a metric space with distance function d and let $(x_n)_{n \in \mathbb{N}}$ be a Cauchy sequence. Then there exists an integer N such that for all m and n greater than N , we have $d(x_m, x_n) < 1$.

Let us fix some $m > N$. For every $i \leq N$, let $d_i = d(x_i, x_m)$. Let $R = \max\{d_1, \dots, d_N, 1\} + 1$. Then, if $i \leq N$, clearly $d(x_m, x_i) < R$. On the other hand, if $i > N$, we have $d(x_m, x_i) < 1$ and so $d(x_m, x_i) < R$. Thus, in any case, $d(x_m, x_i) < R$. This shows that the sequence $(x_n)_{n \in \mathbb{N}}$ is bounded. $\square$

2. COMPLETENESS OF THE REAL NUMBERS

Theorem 7. *The metric space $\mathbb{R}$ is complete.*

Proof. Let $(x_n)_{n \in \mathbb{N}}$ be a Cauchy sequence in $\mathbb{R}$. Then, by Lemma 6, this is a bounded sequence. By the Bolzano-Weierstrass theorem, this sequence has a convergent

subsequence $(x_{n_k})_{k \in \mathbb{N}}$. Let $L = \lim_{k \rightarrow \infty} x_{n_k}$. We will show that $(x_n)_{n \in \mathbb{N}}$ converges to L .

Let $\epsilon > 0$ be given. Since $(x_{n_k})_{k \in \mathbb{N}}$ converges to L , there exists an integer K such that for all $k > K$, we have $|L - x_{n_k}| < \epsilon/2$.

Since $(x_n)_{n \in \mathbb{N}}$ is a Cauchy sequence, there exists an integer N such that for any integers m, n greater than N , we have $|x_m - x_n| < \epsilon/2$. Let J be the smallest integer such that $n_J > N$. Let $M = \max\{K + 1, J\}$.

Now, let n be any integer greater than N . Since $n_M \geq n_J > N$, we have $|x_n - x_{n_M}| < \epsilon/2$. Also, since $M > K$, we have $|L - x_{n_M}| < \epsilon/2$. Thus, by the triangle inequality,

$$|L - x_n| \leq |L - x_{n_M}| + |x_n - x_{n_M}| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Thus, we see that $|L - x_n| < \epsilon$ for any $n > N$. Thus, the sequence $(x_n)_{n \in \mathbb{N}}$ converges to L . $\square$

3. APPLICATION TO CONVERGENCE OF SERIES

Theorem 8 (Cauchy Criterion for convergence of series). *Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. The following two statements are equivalent:*

- (a) *For every $\epsilon > 0$, there exists an integer N such that for any integers m and n satisfying $n > m > N$, we have*

$$|a_{m+1} + \cdots + a_n| < \epsilon.$$

- (b) *The series $\sum_{n=1}^{\infty} a_n$ converges.*

Proof. For any natural number n , let s_n denote the partial sum $s_n := \sum_{k=1}^n a_k$. Then, the series $\sum_{n=1}^{\infty} a_n$ converges if and only if the sequence $(s_n)_{n \in \mathbb{N}}$ converges. This sequence converges if and only if it is a Cauchy sequence.

For any integers m and n such that $m < n$, we have

$$|s_n - s_m| = |a_{m+1} + \cdots + a_n|.$$

Thus, if (a) is true, we see that the sequence $(s_n)_{n \in \mathbb{N}}$ is a Cauchy sequence and so the series $\sum_{n=1}^{\infty} a_n$ converges.

Conversely, if (b) is true, i.e. if the series $\sum_{n=1}^{\infty} a_n$ converges, then the sequence $(s_n)_{n \in \mathbb{N}}$ is a Cauchy sequence. By the above discussion, we see that this implies (a). $\square$

4. READING SUGGESTIONS

The content of this lecture corresponds to Section 2.6 of the text ([1]).

The argument sketched in Remark 5 is written out in detail in [2, Chapter 1]. (Note: Prof. Vasudeva, one of the authors of this book, was a faculty member at IISER Mohali from 2010 to 2016.)

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015.
- [2] Satish Shirali and H. L. Vasudeva, *Metric spaces*. Springer-Verlag London, Ltd., London, 2006. viii+222 pp. 3